# Adebanji Oluwatimileyin Adelowo

Scientific Machine Learning · Mathematical Engineering · Probabilistic Modelling

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## RESEARCH PROFILE

Machine Learning practitioner with an M.Sc. in Mathematical Engineering and a B.Sc. in Mathematics (First Class Honours). Research background in mathematical modelling, numerical methods, optimization, PDEs, and scientific computing, complemented by professional experience developing machine-learning and signal-processing systems for engineering applications.

## RESEARCH INTERESTS

Scientific Machine Learning · Probabilistic Machine Learning · AI for Science · Mathematical Optimization · Uncertainty Quantification · Numerical Analysis · Scientific Computing · Learning for Dynamical Systems · Signal Processing · Data-Efficient Machine Learning

## EDUCATION

2018 – 2021

**M.Sc. Mathematical Engineering** (Final grade: 106/110)
University of L'Aquila, Italy
*Thesis: Bounds on Mixing of Passive Scalars Advected by Incompressible Energy- and Enstrophy-Constrained Flows and Anomalous Dissipation*
*Relevant coursework:* Advanced Optimization; Machine Learning; Dynamical Systems and Bifurcation Theory; Scientific Computing; Functional Analysis; Partial Differential Equations; Fluid Dynamics; Numerical Methods for PDEs; Galerkin / Finite-Element Methods

2014 – 2018

**B.Sc. Mathematics** (First Class Honours, CGPA 4.57/5.0)
Obafemi Awolowo University, Ile-Ife, Nigeria
*Thesis: Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem*
*Relevant coursework:* Real Analysis; Complex Analysis; Topology; Measure Theory; Numerical Analysis; Ordinary Differential Equations; Probability and Statistics

## RESEARCH EXPERIENCE

2020 – 2021

**M.Sc. Thesis Research** | University of L'Aquila, Italy
Research on mathematical bounds and numerical computation for passive-scalar mixing in constrained incompressible flows.

- Derived analytical lower bounds for passive-scalar mixing
- Investigated extremal velocity fields satisfying energy and enstrophy constraints
- Developed a pseudo-spectral numerical solver in Python for computational validation
- Combined mathematical analysis, numerical algorithms, and computational experiments

2017 – 2018

**B.Sc. Thesis Research** | Obafemi Awolowo University, Nigeria
Research on numerical methods for a two-phase free-boundary problem.

- Applied Galerkin finite-element methods to the one-dimensional Stefan problem
- Formulated the weak problem and developed computational algorithms for phase-boundary tracking
- Studied numerical approaches to a physically motivated PDE problem

## PROFESSIONAL EXPERIENCE

Dec 2024 – Present

**Data Scientist** | Intuos Srl, L'Aquila, Italy

- Develop machine-learning models for real-time flight-phase identification from aviation telemetry
- Develop FFT-based signal-processing pipelines for engine RPM estimation and anomaly detection
- Work with high-volume telemetry data and computational pipelines for engineering-system analysis
- Develop REST APIs and operational dashboards integrating machine-learning outputs with engineering data

Apr 2023 – Nov 2024

**Machine Learning Engineer** | N-and Italia Srl, Italy

- Developed predictive machine-learning models for delivery-failure forecasting using large-scale transaction and telemetry data
- Performed statistical analysis of operational and model performance
- Developed data-processing and analytical workflows for operational decision support

Jan 2023 – Mar 2023 — **Operational Data Analyst** | eResult Srl, Cesena, Italy

- Optimised SQL queries for business-intelligence systems
- Automated reporting and analytical workflows
- Improved reproducibility and efficiency of operational data processing

Apr 2022 – Oct 2022 — **Digital Engineering Intern** | Bridgestone EMIA, Rome, Italy

- Improved computational efficiency of half-edge data structures used in tire cross-section finite-element meshing
- Developed geometric algorithms for closest-point computation on cubic Bézier curves

May 2022 – Aug 2022 — **Machine Learning Intern** | Fellowship.AI, San Francisco, USA / Remote

- Implemented an InsetGAN architecture for whole-body human image generation
- Developed and deployed an inference service through FastAPI for integration with a Swift iOS application

## SELECTED RESEARCH & COMPUTATIONAL PROJECTS

### Phase-Field Solver for Boiling Heat Transfer

- Implemented a conservative Allen–Cahn / one-fluid Navier–Stokes phase-field model in Python
- Coupled phase-field, energy, and incompressible Navier–Stokes equations
- Implemented an FFT-based direct solver for the pressure Poisson equation
- Validated against analytical solutions for 2D vapour-bubble growth and the 1D Stefan problem
- Extended the solver to three-dimensional simulations

### Aircraft Engine RPM Estimation via Audio DSP

- Developed a Raspberry Pi-based system for engine RPM estimation from microphone recordings
- Applied FFT-based spectral analysis and frequency-peak detection
- Validated the method using Tecnam P92 / Rotax 912 engine data

### Image Restoration via Total Variation Regularisation

- Implemented TV-regularised variational image restoration
- Formulated the Euler–Lagrange optimality system
- Applied Douglas–Rachford splitting for numerical optimisation

### Deep Image Denoising

- Developed a ResNet architecture with Residual Channel Attention Blocks and Squeeze-and-Excitation
- Investigated blind grayscale denoising with randomly sampled noise levels
- Used SSIM and L1-based training objectives
- Achieved approximately 26–28 dB PSNR on LFW experiments

### LLM Engineering Projects

- Medical Retrieval-Augmented Generation assistant
- Enterprise natural-language-to-SQL system
- Financial sentiment distillation
- Technologies: LangChain, OpenAI, ChromaDB

## TECHNICAL SKILLS

**Programming:** Python | MATLAB | LaTeX
**Machine Learning:** PyTorch | scikit-learn | classification | predictive modelling | deep learning | generative modelling
**Scientific Computing:** NumPy | SciPy | numerical optimization | finite-element methods | pseudo-spectral methods | PDE solvers
**Mathematics:** Mathematical modelling | optimization | numerical analysis | PDEs | dynamical systems | variational methods | probability | statistics
**Signal & Data Processing:** FFT / spectral analysis | telemetry processing | statistical analysis | SQL | Apache Spark | Databricks
**Software & Infrastructure:** FastAPI | REST APIs | Git/GitHub | Kubernetes | MLflow | AWS | Terraform

## TEACHING AND ACADEMIC LEADERSHIP

2016 – 2018 — **Founder & President, Inter-University Mathematics Club** | Nigeria

- Established a cross-university mathematics mentorship network
- Organised seminars on mathematics, research, and research careers

2015 – 2018 — **Academic Director / Peer Tutor, NAMSS** | Obafemi Awolowo University

- Coordinated academic workshops
- Tutored undergraduate students in mathematical analysis and mathematical reasoning

## HONOURS AND SCHOLARSHIPS

**University of L'Aquila International Scholarship** — University of L'Aquila 2018 – 2021

**Professor S.S. Okoya Prize** — Excellence in Applied Mathematics 2018

**MTN Foundation Scholarship** — Merit-based undergraduate scholarship 2014 – 2018

**PTDF Scholarship** — Petroleum Technology Development Fund, Nigeria 2014 – 2018

**Murli T. Chellaram Foundation Scholarship** — Academic Excellence Award 2014 – 2018

## LANGUAGES

English: excellent written and spoken proficiency

## REFERENCES

**Prof. Ogundare Babatunde Sunday**
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Obafemi Awolowo University, Nigeria
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**Dr. Yusuf Tijani**
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University of the Witwatersrand, South Africa
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L'Aquila, August 2026